

Q.No: 6

| DS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Session | Searching | Question Information | Level 1 | Challenge 6 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------|----------------------|---------|-------------|
| <p>Problem Description:</p> <p>Kanna is upset to learn that no one at his school recognises his first name.</p> <p>Even his friends refer to him by his surname.</p> <p>Frustrated, he decides to make his fellow college students know his first name by forcing them to solve this question. The task is determining the third greatest number in the supplied array.</p> <p>Constraints:</p> <p><math>0 &lt; n &lt; 100</math></p> <p><math>0 &lt; arr &lt; 1000</math></p> <p>Input Format:</p> <p>first line represents the number of elements N to be get</p> <p>second line indicates input elements according to N</p> <p>Output Format:</p> <p>Single line represents the out put that is third largest number.</p> |         |           |                      |         |             |

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

6  
1 14 2 16 10 20

EXPECTED OUTPUT

The third Largest element is 14

Test Case 2

INPUT (STDIN)

7  
19 -10 20 14 2 16 10

EXPECTED OUTPUT

The third Largest element is 16

Code:

```
#include <stdio.h>
```

```
void thirdLargest(int arr[], int arr_size)
{
    int first, second, third;

    first = second = third = -1000000;

    for (int i = 0; i < arr_size; i++)
    {
        if (arr[i] > first)
        {
            third = second;
            second = first;
            first = arr[i];
        }
        else if (arr[i] > second)
        {
            third = second;
            second = arr[i];
        }
        else if (arr[i] > third)
        {
            third = arr[i];
        }
    }

    printf("The third Largest element is %d", third);
}

int main()
{
    int n;
    int arr[100];

    scanf("%d", &n);

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    thirdLargest(arr, n);

    return 0;
}
```

Output:

Test\_case 1:

```
aadityajaiswal@Aadityas-MacBook  
6  
1 14 2 16 10 20  
The third Largest element is 14%  
aadityajaiswal@Aadityas-MacBook
```

Test\_case 2:

```
7  
19 -10 20 14 2 16 10  
The third Largest element is 16%  
aadityajaiswal@Aadityas-MacBook Air
```